GreenTech Manufacturing

Operations performance review

Production downtime: where 522 hours went, and how to recover them

Findings from three Power BI dashboards, March to August 2025

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Briefing for executives and clients

EXECUTIVE SUMMARY

Downtime cost 522 production hours, and the main drivers are process and planning, not machines

522 h

Lost to downtime across 645 batches

Batches ran 11.6% over plan (4,515 h planned, 5,037 h actual). 56% of batches were affected.

20.5%

Came from equipment and utilities

The rest traces to sanitation, scheduling, materials and QA holds. Operator-linked factors total 30%.

+23%

No sustained improvement

Lost hours per batch rose from 0.75 h in March to 0.92 h in August.

RECOMMENDED RESPONSE

A 90-day programme with four moves

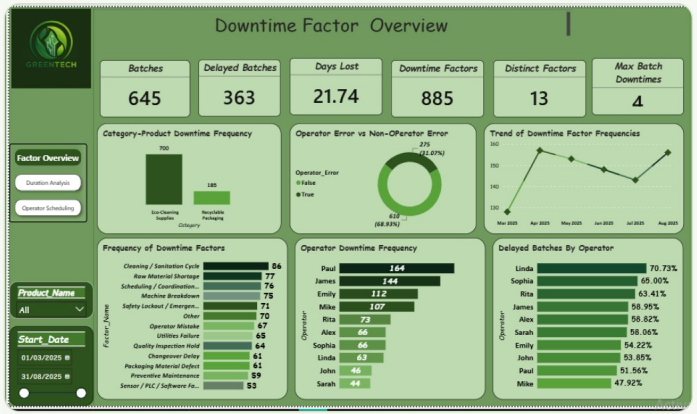
- Time-box sanitation and changeovers
- Secure raw material and packaging supply
- Automate scheduling and rebalance double-batching
- Fix the downtime data and review it monthly

Decision requested: an executive sponsor and two pilot lines, GreenFoam and RePack.

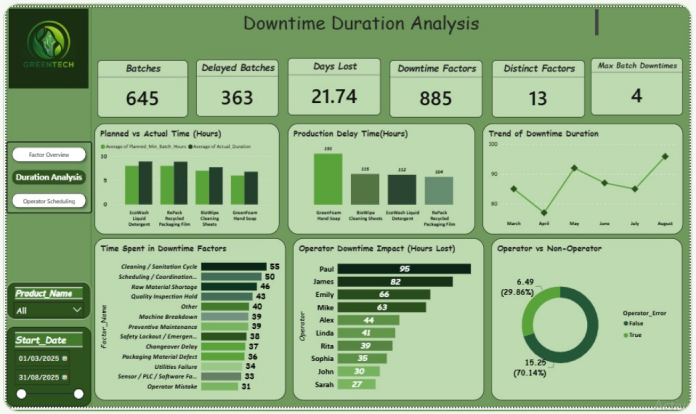
Source: GreenTech batch and downtime data, Mar–Aug 2025 (645 batches, 885 logged factor records)

SCOPE AND APPROACH

We analysed 645 batches and 885 logged downtime records across three dashboards



1 · Factor overview



2 · Duration analysis



3 · Operator scheduling

THE QUESTION

Where, why and how much production time is lost to downtime, and what should we fix first?

THE DATA

Batch records, a downtime log, 13 downtime factors and 4 products. 10 operators, March to August 2025.

DEFINITIONS

Delayed batch: at least one logged stoppage. Days lost: downtime hours ÷ 24.

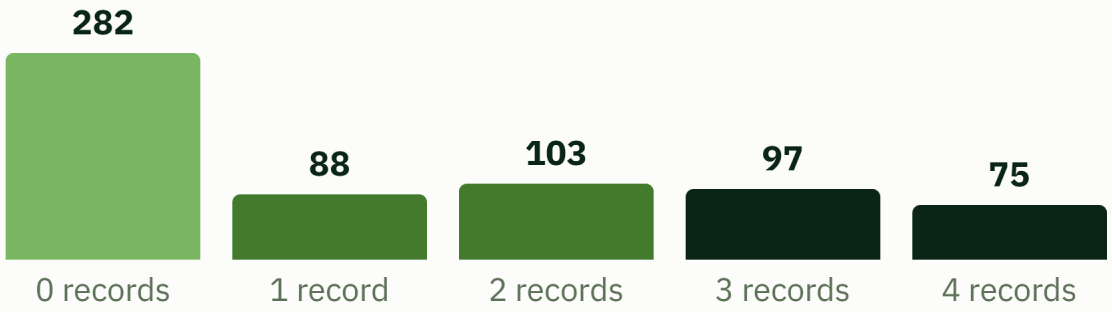
Source: GreenTech Power BI model. 885 logged factor records, of which 875 have positive duration

THE SIZE OF THE PROBLEM

56% of batches were disrupted, and more than one in four had three or four downtime records

Batches	Delayed batches	Downtime records	Production days lost	Distinct factors	Most records in one batch
645	363	885	21.74	13	4
Mar–Aug 2025	56% of all batches	875 above zero minutes	522 hours	No single driver	75 batches

Batches by number of downtime records



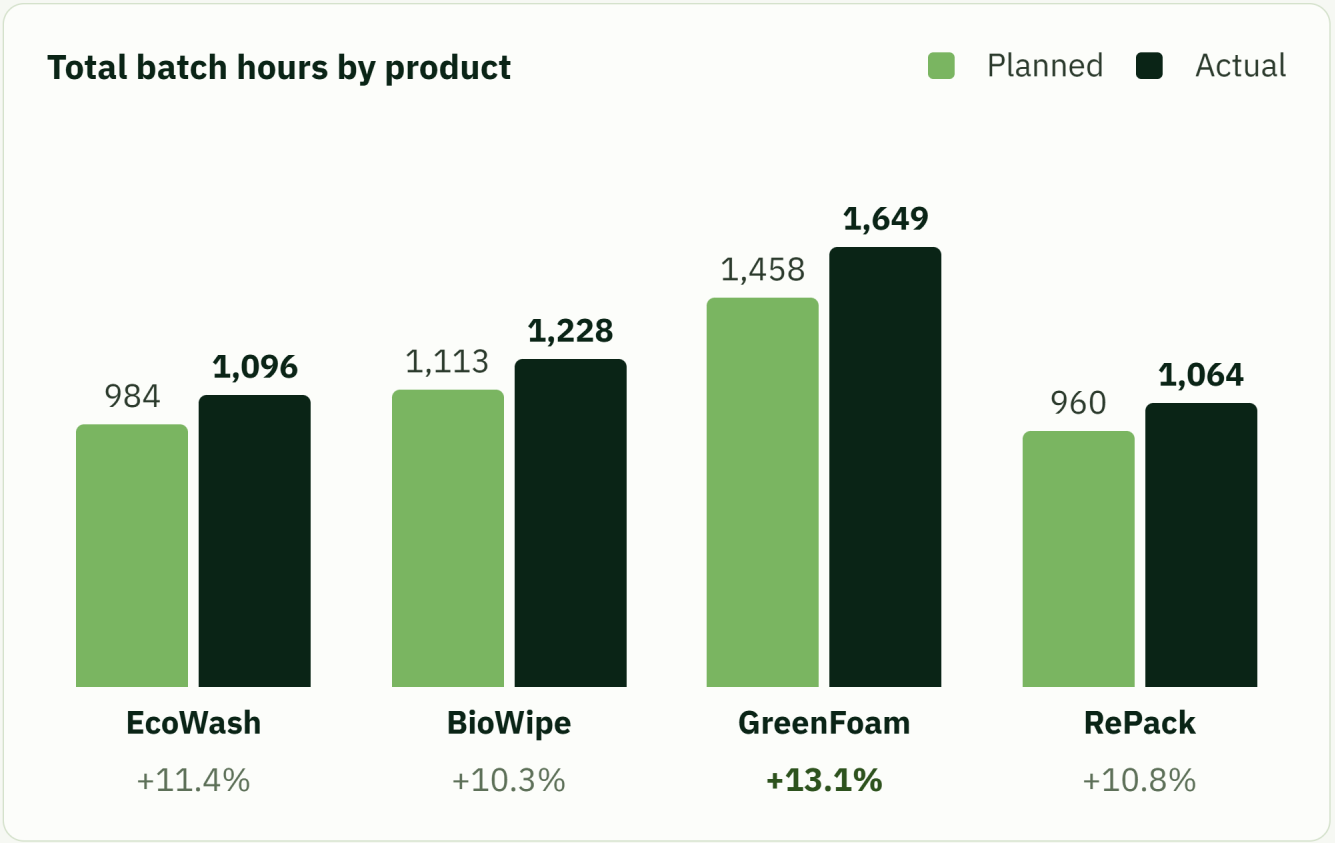
SO WHAT

The average record lasts about 35 minutes and none exceeded two hours. The loss comes from frequency and repetition: 172 batches had three or four downtime records.

Source: GreenTech downtime data, Mar–Aug 2025. 885 records, 10 of them zero-minute. Days lost = 522 h ÷ 24

FINDING 1 · PLAN VERSUS ACTUAL

Logged downtime explains virtually all of the overrun: 4,515 planned hours became 5,037 actual hours



282 of 645

Batches had no logged downtime and finished exactly on plan.

522.0 ≈ 521.8

The 522-hour schedule gap and 521.8 logged hours differ by only 10 minutes.

SO WHAT

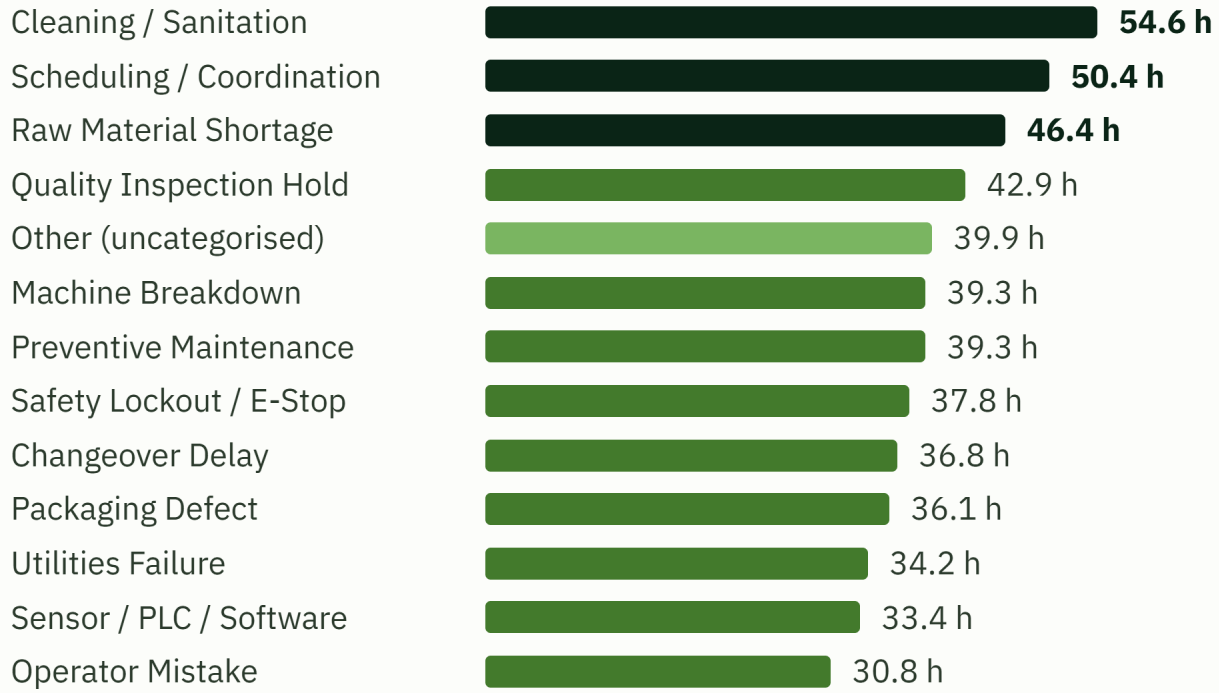
Nearly every hour recovered from downtime comes straight back to the schedule.

Source: GreenTech batch and downtime data, Mar–Aug 2025. Percentages = overrun ÷ planned hours

FINDING 2 · WHERE THE HOURS WENT

No single culprit: it takes nine of 13 factors to account for 70% of lost hours

Hours lost by downtime factor



29%

of lost hours sit in the top three: cleaning, scheduling and raw materials.

20.5%

come from equipment and utilities combined. Machines are not the main story.

7.6%

sits in ‘Other’, the fifth-largest category. That is a data gap we can close.

Source: GreenTech downtime log, Mar–Aug 2025. Darkest bars: top three factors

FINDING 3 · PROCESS VERSUS PEOPLE

Process, supply and equipment account for 70% of lost hours; operator-linked factors for 30%

Share of downtime by cause type

Non-operator Operator-linked

By logged records (885)



By hours lost (522 h)



Read with care: the model's 'operator-linked' flag also covers scheduling, changeover and safety lockouts. Pure operator mistakes are 30.8 h, 5.9% of the total.

Four buckets of lost hours

Human and coordination — **155.8 h · 29.9%**

Scheduling, changeover, operator mistake, safety lockout

Sanitation, maintenance and QA — **136.7 h · 26.2%**

Cleaning, preventive maintenance, quality holds

Materials and uncategorised — **122.4 h · 23.5%**

Raw material shortage, packaging defects, other

Equipment and utilities — **106.9 h · 20.5%**

Machine breakdown, utilities failure, PLC and software

Source: GreenTech downtime log and factor table (Operator_Error flag), Mar–Aug 2025

FINDING 4 · PRODUCTS

GreenFoam carries the volume, but RePack and EcoWash lose more per batch

Product	Batches	Incidents	Hours lost	Incidents per batch	Hours lost per batch	Over plan
EcoWash Liquid Detergent	123	173	112.0	1.41	0.91	+11.4%
BioWipe Cleaning Sheets	159	204	114.9	1.28	0.72	+10.3%
GreenFoam Hand Soap	243	323	190.9	1.33	0.79	+13.1%
RePack Recycled Packaging Film	120	185	104.1	1.54	0.87	+10.8%

Volume explains the lead

GreenFoam is 38% of batches and 37% of lost hours. Its rate per batch (1.33) is below the 1.37 average.

Category totals mislead

Eco-Cleaning shows 700 incidents against 185, but over 525 batches versus 120: 1.33 against 1.54 per batch.

Short batches feel it most

GreenFoam has the shortest planned batch (6 h), so stoppages cost it +13.1% against +10.3% to +11.4% elsewhere.

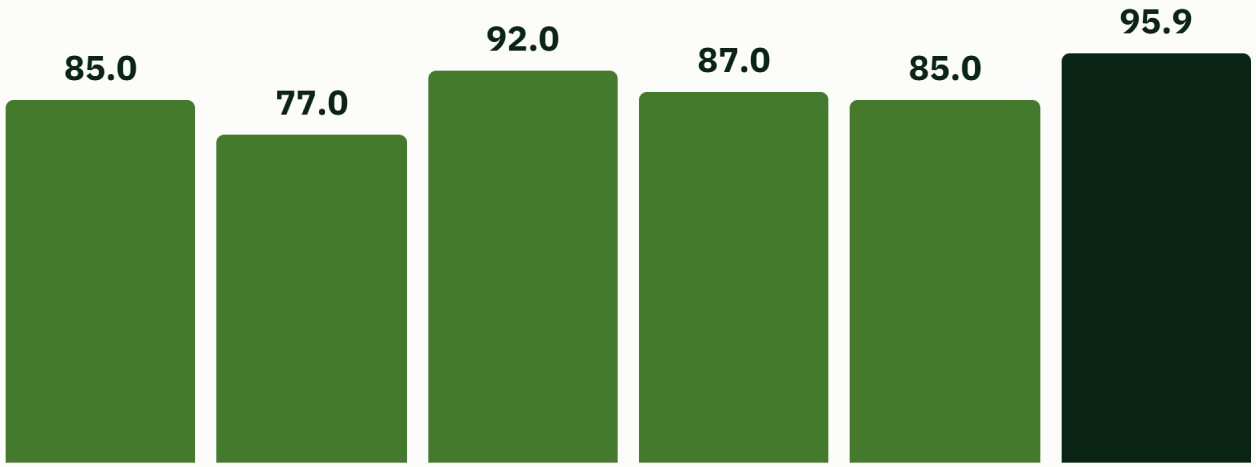
Source: GreenTech batch and downtime data, Mar–Aug 2025. Line average: 1.37 incidents per batch

FINDING 5 · TREND

Lost hours are not falling: per batch, they rose from 0.75 h in March to 0.92 h in August

Downtime hours by month

Under each month: hours lost per batch



+23%

Lost hours per batch, March to August, with batch volumes steady at 100 to 114 a month.

87 h

Average lost every month. No month fell below 77 h.

157

Incidents in April, the peak count, yet its hours were the lowest. Count and duration move separately.

Source: GreenTech batch and downtime data, Mar–Aug 2025. Hours per batch = downtime hours ÷ batches started in month

FINDING 6 · OPERATORS

Raw counts point at Paul and James; per batch worked, Rita, Sophia and Linda stand out

Operator	Batches	Incidents	Per batch	Delayed
Rita	41	73	1.78	63%
Sophia	40	66	1.65	65%
Linda	41	63	1.54	71%
James	95	144	1.52	59%
Sarah	31	44	1.42	58%
Emily	83	112	1.35	54%
Alex	51	66	1.29	59%
Paul	128	164	1.28	52%
John	39	46	1.18	54%
Mike	96	107	1.11	48%

Volume explains Paul's lead

He ran 128 batches (20% of the total) and logged 164 incidents (19%). At 1.28 per batch he is below the 1.37 average.

Three operators sit above the line

Rita, Sophia and Linda run 1.54 to 1.78 incidents per batch and 63% to 71% delayed batches, against 56% overall.

A prompt, not a verdict

Operators worked 31 to 128 batches each. Use these rates to ask about workload and conditions.

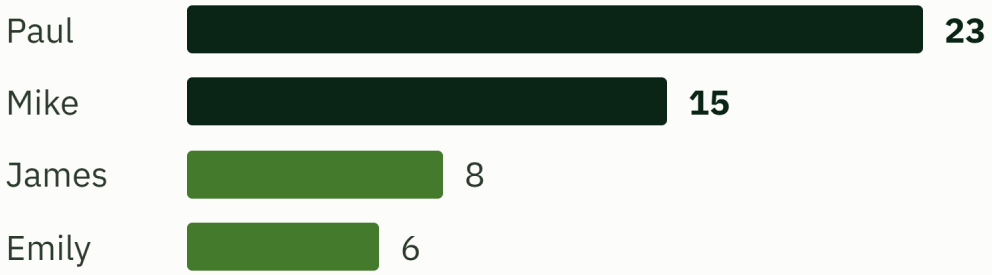
Source: GreenTech batch and downtime data, Mar–Aug 2025. Sorted by incidents per batch worked

FINDING 7 · SCHEDULING

Scheduling is the second-largest loss, and double-batching is concentrated in Paul and Mike

Days an operator ran two or more batches

52 operator-days across 51 production days



The other six operators: none

50.4 h — Lost to scheduling and coordination: 76 incidents.

109 — Batches were run on double-batch days, with 131 incidents.

1.20 — Incidents per batch on double-batch days, against 1.41 on other days.

SO WHAT Double-batching is not itself raising stoppages. The cost sits in coordination and handoffs, so fix the plan and the schedule, not the person.

Source: GreenTech batch and downtime data, Mar–Aug 2025 (Power BI page: Operator Scheduling)

RECOMMENDATIONS · FROM THE PROJECT REVIEW

Nine actions from the review, each tied to evidence in the data

Type	Recommendation	Evidence
Operational	Optimise cleaning and sanitation schedules	Largest factor: 86 incidents, 54.6 h
Operational	Strengthen raw material planning and stock monitoring	77 incidents, 46.4 h lost
Operational	Automate production scheduling where possible	76 incidents, 50.4 h; second-largest loss
Operational	Reduce overlapping batch assignments	51 double-batch days; 73% Paul and Mike
Operational	Investigate recurring high-rate patterns; coach after weighing workload and product mix	Highest rates: Rita, Sophia, Linda (1.54–1.78)
Operational	Plan preventive maintenance for high-risk equipment	Breakdowns, utilities, PLC faults: 107 h
Strategic	Real-time downtime monitoring dashboards	Builds on the Power BI model in place
Strategic	Monthly downtime reviews of recurring bottlenecks	Flat trend: 77 to 96 h lost every month
Strategic	Operator scorecards for continuous improvement	Measure per batch worked, not raw counts

Source: project review recommendations; evidence from GreenTech batch and downtime data, Mar–Aug 2025

RECOMMENDATIONS · OUR ADDITIONS

Five additions that target the largest pools of lost hours

Addition	What we would do	Hours in play
A · Time-box sanitation and changeovers	Set a standard time per product and clean type. Cleaning stays mandatory, so shorten it rather than skip it.	91 h · 17.5%
B · Gate incoming materials	Minimum-stock triggers for raw materials, incoming checks on packaging and supplier scorecards.	82.5 h · 15.8%
C · Fix the downtime data	‘Other’ is the fifth-largest factor. Make reason codes mandatory and log start and end times for every stop.	39.9 h · 7.6%
D · Run product reliability sprints	Start with RePack (1.54 incidents per batch) and EcoWash (0.91 h lost per batch) to find product-specific causes.	216 h · 41.4%
E · Agree a QA hold protocol	Pre-agree sampling plans and release criteria with QA. Holds average 40 minutes, the longest of any factor.	42.9 h · 8.2%

NOTE Hours in play are the 2025 hours each action addresses. They overlap, so they are not additive and are not savings forecasts.

Source: Consultant analysis of GreenTech batch and downtime data, Mar–Aug 2025

ROADMAP

A 90-day plan: baseline the data, pilot the fixes, then embed the review rhythm

Days 1–30 · Baseline

- Mandatory reason codes; retire ‘Other’
- Automate the Power BI refresh
- Agree targets by factor with operations, QA and supply

Days 31–60 · Pilot

- Time-box sanitation and changeovers on GreenFoam and RePack
- Stock triggers for top raw materials
- Trial automated scheduling rules; rebalance double-batch days

Days 61–90 · Embed

- Monthly downtime review with an owner per factor
- Scorecards per batch worked, used for coaching
- Report recovered hours against the 522-hour baseline

≈130 h

ILLUSTRATIVE PRIZE

Recovering 25% of lost hours is approximately 19 more standard batches over six months. A scenario, not a forecast.

BASELINES TO BEAT

0.81 h lost per batch · 56% of batches delayed · 7.6% of hours uncategorised

Source: Consultant analysis. 130 h ÷ 7.0 h average planned batch = 18.6, about 19 batches

DECISIONS REQUESTED

The data explains every lost hour. The next step is ownership.

1

Name an executive sponsor

Owns the 522-hour baseline and chairs the monthly downtime review.

2

Approve the data fixes

Mandatory reason codes and an automated refresh of the Power BI model.

3

Choose the pilot lines

GreenFoam (highest volume, +13.1% over plan) and RePack (1.54 incidents per batch).